

UART Protocol - Parameter ID's (FOC v4.4.2)

Green highlight indicates parameters that are normally used while controlling the motor. When loading values using hardcoded default settings in firmware, it is usually not necessary to use parameters other than the ones highlighted in green.

PARM Code (Dec)	PARM Code (Hex)	Parameter Name	Category	R/W	Description	Units	Data Format
1	1	status_motor	Status	R	Motor status flag	None	uint32
2	2	fault_warning	Status	R	Motor fault & warning flag	None	uint32
3	3	current_limit	FOC Config	R/W	HP DAC value for over-current limit	DAC(8b)	uint32
4	4	dt	FOC Config	R/W	FOC control period	sec * 1024 / 2^16	q16
5	5	est_angle_offset	Obsolete	-	-	-	-
6	6	open_loop_align_hold_num_steps	Align&Go Config	R/W	Align hold time	dt counts	uint32
7	7	open_loop_speed_slew_rate	Align&Go Config	R/W	Open loop speed ramp rate	rad/dt/dt	q16
8	8	open_loop_speed_setpoint	Align&Go Config	R/W	Open loop speed target	rad/dt	q16
9	9	closed_loop_speed_slew_rate	Outerloop Control	R/W	Closed loop speed ramp rate	rad/dt/dt	q16
10	A	closed_loop_speed_setpoint	Operation	R/W	Closed loop speed target	rad/dt	q16
11	B	id_ref	Align&Go Config	R/W	Open loop id target	Amps/ current_descale	q16
12	C	iq_ref	Align&Go Config	R/W	Open loop iq target	Amps/ current_descale	q16
13	D	id_pi_filter_coeff	Current Control	R/W	Coefficient for id filter	None	q16
14	E	iq_pi_filter_coeff	Current Control	R/W	Coefficient for iq filter	None	q16
15	F	id_fdbk	Obsolete	-	-	-	-
16	10	iq_fdbk	Obsolete	-	-	-	-
17	11	Vdc_gui	Obsolete	-	-	-	-
18	12	speed_pid_Kp	Outerloop Control	R/W	Proportional gain for Speed PI controller	None	q16
19	13	speed_pid_Ki	Outerloop Control	R/W	Integral gain for Speed PI controller	None	q16
20	14	outerloop_pid_min	Outerloop Control	R/W	Min saturation limit for speed PI controller	Amps/ current_descale	q16
21	15	outerloop_pid_max	Outerloop Control	R/W	Max saturation limit for speed PI controller	Amps/ current_descale	q16
22	16	Id_pid_Kp	Current Control	R/W	Proportional gain for id PI controller	None	q16
23	17	Id_pid_Ki	Current Control	R/W	Integral gain for id PI controller	None	q16
24	18	Id_pid_min	Current Control	R/W	Min saturation limit for id PI controller	Amps/ current_descale	q16
25	19	Id_pid_max	Current Control	R/W	Max saturation limit for id PI controller	Amps/ current_descale	q16
26	1A	Iq_pid_Kp	Current Control	R/W	Proportional gain for iq PI controller	None	q16

27	1B	Iq_pid_Ki	Current Control	R/W	Integral gain for iq PI controller	None	q16
28	1C	Iq_pid_min	Current Control	R/W	Min saturation limit for iq PI controller	Amps/ current_descale	q16
29	1D	Iq_pid_max	Current Control	R/W	Max saturation limit for iq PI controller	Amps/ current_descale	q16
30	1E	Ld	Motor Config	R/W	Motor d-axis phase inductance	uH	q26
31	1F	Lq	Motor Config	R/W	Motor q-axis phase inductance	uH	q26
32	20	Ralpha	Motor Config	R/W	Motor phase winding resistance	Ohms	q16
33	21	num_pole_pairs	Motor Config	R/W	Number of motor pole pairs	None	q16
34	22	Config_Ke	Motor Config	R/W	Motor BEMF constant	V*sec/rad	q26
35	23	Confi_J	Motor Config	R/W	Motor load inertia constant	Nm/a	q26
36	24	NominalSpeed	Motor Config	R/W	Motor nominal speed	rad/s	q16
37	25	estimated_angle	Status	R	Estimated angle	rad	q16
38	26	estimated_speed	Status	R	Estimated speed	rad/second	q16
39	27	ideal_angle	Obsolete	-	-	-	-
40	28	measured_speed	Obsolete	-	-	-	-
41	29	motor_current	Obsolete	-	-	-	-
41	29	pwm_dac_selection	Obsolete	-	-	-	-
43	2B	debug1	Debug	R	Debug output variable	None	q16
44	2C	debug2	Debug	R	Debug output variable	None	q16
45	2D	debug3	Debug	R	Debug output variable	None	q16
46	2E	debug4	Debug	R	Debug output variable	None	q16
47	2F	debug5	Debug	R	Debug output variable	None	q16
48	30	debug6	Debug	R	Debug output variable	None	q16
49	31	scale_factor1	Debug	W	Scale factor for PWM DAC #1	None	q16
50	32	scale_factor2	Debug	W	Scale factor for PWM DAC #2	None	q16
51	33	scale_factor3	Debug	W	Scale factor for PWM DAC #3	None	q16
52	34	debug1_index	Debug	W	Index of desired signal to view on Debug #1	None	uint32
53	35	debug2_index	Debug	W	Index of desired signal to view on Debug #2	None	uint32
54	36	debug3_index	Debug	W	Index of desired signal to view on Debug #3	None	uint32
55	37	debug1_min	Debug	R	Debug#1 minimum value	None	q16
56	38	debug1_max	Debug	R	Debug#1 maximum value	None	q16
57	39	debug2_min	Debug	R	Debug#2 minimum value	None	q16
58	3A	debug2_max	Debug	R	Debug#2 maximum value	None	q16
59	3B	debug3_min	Debug	R	Debug#3 minimum value	None	q16
60	3C	debug3_max	Debug	R	Debug#3 maximum value	None	q16
61	3D	id_ref_cl	Operation	R/W	Closed loop id target	Amps/ current_descale	q16

62	3E	idrq_ramp_rate	Current Control	R/W	Ramp rate for id/iq reference	Amps/ current_descale/ FOC_control_perio d	q16
63	3F	torque_scale	Obsolete	-	-	-	-
64	40	torque_offset	Obsolete	-	-	-	-
65	41	torque_angle_offset	Obsolete	-	-	-	-
66	42	descale_current_factor	Hardware Config	R	Current descale factor	None	q16
67	43	est_lock_assist_vq	Obsolete	-	-	-	-
68	44	cl_mode_selection	Operation	R/W	Flag: 0=TorqueControl / 1=SpeedControl	None	uint32
69	45	iq_ref_cl	Operation	R/W	Closed loop iq target (Torque control only)	Amps/ current_descale	q16
70	46	reverse_direction	Operation	R/W	Flag: 0=Forward / 1=Reverse	None	uint32
71	47	cl_outerloop_setpoint_ramp	Obsolete	-	-	-	-
72	48	misc_q16_1	Debug	W	Debug input variable	None	q16
73	49	misc_q16_2	Debug	W	Debug input variable	None	q16
74	4A	misc_q16_3	Debug	W	Debug input variable	None	q16
75	4B	misc_uint32_1	Debug	W	Debug input variable	None	uint32
76	4C	misc_uint32_2	Debug	W	Debug input variable	None	uint32
77	4D	misc_uint32_3	Debug	W	Debug input variable	None	uint32
78	4E	PWM_to_FOC_control_ratio	FOC Config	R/W	Ratio of PWM cycles per FOC control period	PWM_cycles/ FOC_control_perio d	uint32
79	4F	diff_amp_gain	Hardware Config	R/W	Current sensing diff amp gain	None	uint32
80	50	Motor_BEMF_const	Stall Detection	R/W	Motor BEMF constant (for stall detection)	Volts/rad/s	q16
81	51	Stall_Detection_TimeOut	Stall Detection	R/W	Stall detection blanking time	SysTick timer counts	uint32
82	52	Auto_Mode_Open_Close_Loop	Align&Go Config	R/W	Closed loop switch mode: 0=Manual / 1=Auto	None	uint32
83	53	Clear_Fault_Flags	Status	W	Clear fault command: 1=Clear	None	uint32
84	54	fw_id_sp	Field Weakening	R/W	Maximum id reference limit during field weakening	Amps/ current_descale	q16
85	55	fw_vdrive_threshold	Obsolete	-	-	-	-
86	56	fw_speed_threshold_sp	Field Weakening	R/W	Minimum speed at which field weakening is allowed	rad/s	q16
87	57	param_read_command	Debug	W	Debug export index	None	int32
88	58	param1	Debug	R	Debug export value	None	int32
89	59	param2	Debug	R	Debug export value	None	int32
90	5A	param3	Debug	R	Debug export value	None	int32
91	5B	ipd_charge	IPD	R/W	Charging duty cycle	PWM timer ticks	uint32

92	5C	ipd_discharge	IPD	R/W	Discharging duty cycle as a percentage of charging duty cycle	%	uint32
93	5D	ipd_num_periods_per_pulse	IPD	R/W	Number of PWM periods per pulse	None	uint32
94	5E	ipd_angle_offset_forward	IPD	R/W	Angle offset in forward direction	rad	int32
95	5F	ipd_angle_offset_reverse	IPD	R/W	Angle offset in reverse direction	rad	int32
96	60	ipd_same_pol_diff	IPD	R	Current delta between the top 2 phase currents	ADC counts	int32
97	61	ipd_max_min_diff	IPD	R	Current delta between the min and max phase currents	ADC counts	int32
98	62	ipd_angle	IPD	R	Calculated IPD angle	rad	int32
99	63	sim_braking_align	SIM	R/W	Align time when SIM detects motion before start	control periods	uint32
100	64	sim_braking_ol_id_iq_ref_mul	SIM	R/W	Id reference braking multiplier when SIM detects motion before start	None	q16
101	65	sim_braking_ol_iq_ref_mul	SIM	R/W	Iq reference braking multiplier when SIM detects motion before start	None	q16
102	66	min_cl_speed_limit	Obsolete	-	-	-	-
103	67	max_cl_speed_limit	App Config	R/W	Maximum speed target that the motor is allowed to reach	rad/s	q16
104	68	sim_braking_timeout	SIM	R/W	SIM timeout	control periods	uint32
105	69	sim_braking_diff_threshold	Obsolete	-	-	-	-
106	6A	sim_current_track_gain	Obsolete	-	-	-	-
107	6B	sim_bemf_amp_gain	SIM	R/W	BEMF ADC channel amplifier gain	BEMF amplifier gain index	q16
108	6C	close_loop_coast_timeout	App Config	R/W	Time to timeout during coasting if enabled	DivSystick counts	uint32
109	6D	pwm_dac_selection1	Debug	W	Signal selection index for PWM DAC #1	None	uint32
110	6E	pwm_dac_selection2	Debug	W	Signal selection index for PWM DAC #2	None	uint32
111	6F	pwm_dac_selection3	Debug	W	Signal selection index for PWM DAC #3	None	uint32
112	70	pwm_dac_selection4	Debug	W	Signal selection index for PWM DAC #4	None	uint32
113	71	scale_factor4	Debug	W	Scale factor for PWM DAC #4	None	q16
114	72	Phase_Rsense	Hardware Config	R	Current sense resistor value for motor control	Ohms	q16
115	73	OC_Rsense	Hardware Config	R	Current sense resistor value for over-current protection	Ohms	q16
116	74	Power_pid_Kp	Outerloop Control	R/W	Proportional gain for Power PI controller	None	q16
117	75	Power_pid_Ki	Outerloop Control	R/W	Integral gain for Power PI controller	None	q16
118	76	Power_ramp_rate	Outerloop Control	R/W	Power ramp rate	Watts/dt	q16
119	77	Power_setpoint	Operation	R/W	Power target	Watts	q16
120	78	Power_max_limit	App Config	R/W	Maximum power target that the motor is allowed to reach	Watts	q16
121	79	SMO_Kpl	Obsolete	-	-	-	-
122	7A	SMO_Kangle	Obsolete	-	-	-	-

123	7B	SMO_Kslide	Obsolete	-	-	-	-
124	7C	SMO_Kslf	Obsolete	-	-	-	-
125	7D	SMO_Kspeed	Obsolete	-	-	-	-
126	7E	LBG1_Kpe	MPOS-Estimator	R	Luenberger observer error gain 1	None	q16
127	7F	LBG1_Kpi	MPOS-Estimator	R	Luenberger observer error gain 2	None	q16
128	80	LBG1_BandWidth	MPOS-Estimator	R/W	Luenberger observer bandwidth	rad/s	q16
129	81	LBG1_DampingFactor	MPOS-Estimator	R/W	Luenberger observer damping factor	None	q16
130	82	LBG1_fc_speed	MPOS-Estimator	R/W	Luenberger observer speed filter cutoff frequency	Hz	q16
131	83	LBG2_Kpl	Obsolete	-	-	-	-
132	84	LBG2_KpE	Obsolete	-	-	-	-
133	85	LBG2_Kspeed	Obsolete	-	-	-	-
134	86	angle_correction_offset	MPOS-Estimator	R/W	Luenberger observer angle correction offset	rad	q16
135	87	angle_correction_factor	MPOS-Estimator	R/W	Luenberger observer angle correction gain (speed dependent)	s	q20
136	88	firmware_options	Status	R	Used by GUI to read which features and modes are enabled in the firmware	None	uint32
137	89	calculated_power	Status	R	Calculated motor power	Watts	q16
138	8A	Vdc_Rsense_ratio	Hardware Config	R	Resistor divider ratio for DC bus voltage measurement (R1+R2)/R2	None	q16
139	8B	Vphase_Rsense_ratio	Hardware Config	R	Resistor divider ratio for phase voltage measurement (R1+R2)/R2	None	q16
140	8C	Vdc_max_limit	Hardware Config	R	Maximum DC bus voltage allowed for the selected PCB	Volts	q16
141	8D	AutoTune_Kpv	Obsolete	-	-	-	-
142	8E	AutoTune_Kiv	Obsolete	-	-	-	-
143	8F	ATune_Command	Autotuning	W	Autotuning command	None	uint32
144	90	ATune_MID2_Istep	Autotuning	R/W	MID2 iq reference step size	% current scale	q16
145	91	ATune_ATPI_Delta	Autotuning	R/W	ATPI stability index for speed controller	None	q6
146	92	AutoTune_Status	Autotuning	R	Autotuning status message	None	uint32
147	93	ATune_MID2_RunMode	Autotuning	W	MID2 run mode selection	None	uint32
148	94	AutoTune_Storage_ke_q26	Autotuning	R	Measured Ke BEMF constant	V*sec/rad	q26
149	95	AutoTune_Storage_j_q26	Autotuning	R	Measured J motor load inertia	Nm/a	q26
150	96	AutoTune_Storage_speed_ka_q16	Autotuning	R	Calculated speed controller Ka gain	None	q16
151	97	AutoTune_Storage_speed_kb_q16	Autotuning	R	Calculated speed controller Kb gain	None	q16
152	98	AutoTune_Storage_id_ka_q16	Autotuning	R	Calculated id controller Ka gain	None	q16
153	99	AutoTune_Storage_id_kb_q16	Autotuning	R	Calculated id controller Kb gain	None	q16
154	9A	AutoTune_Storage_iq_ka_q16	Autotuning	R	Calculated iq controller Ka gain	None	q16
155	9B	AutoTune_Storage_iq_kb_q16	Autotuning	R	Calculated iq controller Kb gain	None	q16
156	9C	ATune_MID1_Start	Autotuning	W	MID1 start button command	None	uint32

157	9D	ATune_MID1_RunMode	Autotuning	W	MID1 run mode selection	None	uint32
158	9E	AutoTune_MID1_Storage_R_q16	Autotuning	R	Measured phase resistance	Ohms	q16
159	9F	AutoTune_MID1_Storage_Ld_q26	Autotuning	R	Measured d-axis phase inductance	uH	q26
160	A0	AutoTune_MID1_Storage_Lq_q26	Autotuning	R	Measured q-axis phase inductance	uH	q26
161	A1	ATune_MID1_AlignV1	Autotuning	R/W	MID1 voltage level #1	% DC bus voltage/100	q16
162	A2	ATune_MID1_AlignV2	Autotuning	R/W	MID1 voltage level #2	% DC bus voltage/100	q16
163	A3	ATune_MID1_Vinj	Autotuning	R/W	MID1 injection voltage amplitude	% DC bus voltage/100	q16
164	A4	ATune_MID1_numlimit	Autotuning	R	MID1 control periods to complete an injection waveform period	control periods	uint32
165	A5	ATune_MID1_ManCmd	Autotuning	W	MID1 manual mode command	None	uint32
166	A6	ATune_MID1_PFactor	Autotuning	R	Square value of measured power factor	None	q16
167	A7	Commutation_Mode	Status	R	Indicates MPOS mode selected in firmware	None	uint32
168	A8	HallConfig_Istart	MPOS-Hall	R/W	Starting current target for hall sensor configuration	Amps/current_descale	q16
169	A9	HallConfig_Speed	MPOS-Hall	R/W	Speed switchover setting between hall mode and estimator mode	rad/s	q16
170	AA	HallConfig_Gain_Fwd	MPOS-Hall	R/W	Hall sensor angle correction gain in forward direction (speed dependent)	s	q20
171	AB	HallConfig_Gain_Rev	MPOS-Hall	R/W	Hall sensor angle correction gain in reverse direction (speed dependent)	s	q20
172	AC	HallConfig_Offset_Fwd	MPOS-Hall	R/W	Hall sensor angle correction offset in forward direction	rad	q16
173	AD	HallConfig_Offset_Rev	MPOS-Hall	R/W	Hall sensor angle correction offset in reverse direction	rad	q16
174	AE	HallConfig_Snap_1_Fwd	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #1 in forward direction	rad	q16
175	AF	HallConfig_Snap_1_Rev	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #1 in reverse direction	rad	q16
176	B0	HallConfig_Snap_2_Fwd	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #2 in forward direction	rad	q16
177	B1	HallConfig_Snap_2_Rev	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #2 in reverse direction	rad	q16
178	B2	HallConfig_Snap_3_Fwd	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #3 in forward direction	rad	q16
179	B3	HallConfig_Snap_3_Rev	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #3 in reverse direction	rad	q16

180	B4	HallConfig_Snap_4_Fwd	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #4 in forward direction	rad	q16
181	B5	HallConfig_Snap_4_Rev	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #4 in reverse direction	rad	q16
182	B6	HallConfig_Snap_5_Fwd	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #5 in forward direction	rad	q16
183	B7	HallConfig_Snap_5_Rev	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #5 in reverse direction	rad	q16
184	B8	HallConfig_Snap_6_Fwd	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #6 in forward direction	rad	q16
185	B9	HallConfig_Snap_6_Rev	MPOS-Hall	R/W	Hall angle assigned from hall angle table to state #6 in reverse direction	rad	q16
186	BA	HallTuning_Status	Hall Tuning	R	Hall tuning status message	None	uint32
187	BB	HallTuning_Command	Hall Tuning	W	Hall tuning command	None	uint32
188	BC	HallTuning_Istart	Hall Tuning	R/W	Align & go current to use during hall tuning	Amps/ current_descale	q16
189	BD	HallTuning_Speed_Ramp	Hall Tuning	R/W	Align & go speed ramp rate to use during hall tuning	rad/dt/dt	q16
190	BE	HallTuning_Target_Speed	Hall Tuning	R/W	Align & go speed target to use during hall tuning	rad/dt	q16
191	BF	HallTuning_State_1	Hall Tuning	R	Forced angle reading during hall tuning for state #1	rad	q16
192	C0	HallTuning_Snap_1	Hall Tuning	R	Hall angle table assignment during hall tuning for state #1	rad	q16
193	C1	HallTuning_State_2	Hall Tuning	R	Forced angle reading during hall tuning for state #2	rad	q16
194	C2	HallTuning_Snap_2	Hall Tuning	R	Hall angle table assignment during hall tuning for state #2	rad	q16
195	C3	HallTuning_State_3	Hall Tuning	R	Forced angle reading during hall tuning for state #3	rad	q16
196	C4	HallTuning_Snap_3	Hall Tuning	R	Hall angle table assignment during hall tuning for state #3	rad	q16
197	C5	HallTuning_State_4	Hall Tuning	R	Forced angle reading during hall tuning for state #4	rad	q16
198	C6	HallTuning_Snap_4	Hall Tuning	R	Hall angle table assignment during hall tuning for state #4	rad	q16
199	C7	HallTuning_State_5	Hall Tuning	R	Forced angle reading during hall tuning for state #5	rad	q16
200	C8	HallTuning_Snap_5	Hall Tuning	R	Hall angle table assignment during hall tuning for state #5	rad	q16

201	C9	HallTuning_State_6	Hall Tuning	R	Forced angle reading during hall tuning for state #6	rad	q16
202	CA	HallTuning_Snap_6	Hall Tuning	R	Hall angle table assignment during hall tuning for state #6	rad	q16
203	CB	QEP_Resolution	MPOS-QEP	R	QEP resolution (non-quadrature)	Counts/rev	uint32
204	CC	QEP_Offset_Angle_Fwd	MPOS-QEP	R/W	QEP angle correction offset in forward direction	rad	q16
205	CD	QEP_Offset_Angle_Rev	MPOS-QEP	R/W	QEP angle correction offset in reverse direction	rad	q16
206	CE	QEP_Fc	MPOS-QEP	R/W	QEP speed filter cutoff frequency	Hz	q16
207	CF	mtpa_gain	MTPA	R/W	Gain used in calculating the MTPA angle	Deg/Amp	q16
208	D0	capture_mode	Debug	R/W	Select how many control cycles to fetch data	None	uint32
209	D1	capture_size	Debug	R/W	Configure the sample size in the device	None	uint32
210	D2	capture_param_selection	Debug	R/W	Select which variables to be captured like DAC/DAS	None	uint32
211	D3	capture_buffer_index	Debug	R	Select the buffer index	None	uint32
212	D4	capture_buffer_value_at_index	Debug	R	Read the value at the buffer index	None	uint32
213	D5	capture_start	Debug	R	Send the command to start capture	None	uint32
214	D6	capture_status	Debug	R	Read the status of data capture	None	uint32
215	D7	capture_cnt	Debug	R	Check the number of data capured	None	uint32
216	D8	FLASH_read	Flash Operation	W	Button to trigger read data from Flash	None	uint32
217	D9	FLASH_write	Flash Operation	W	Button to trigger write data to Flash	None	uint32
218	DA	FLASH_erase	Flash Operation	W	Button to trigger erase data in Flash	None	uint32
219	DB	Forced_angle_mode	Angle Forcing	W	Command for angle forcing	None	uint32
220	DC	Forced_angle	Angle Forcing	R/W	Forced angle (Electrical)	Degree	uint32
221	DD	Forced_angle_current	Angle Forcing	R/W	Current command during angle forcing	Amp	q16
222	DE	Forced_angle_speed	Angle Forcing	R/W	Speed limit during angle forcing	Hz	q16
223	DF	QEP_Offset_CalEnable	MPOS-QEP	W	Command to enable QEP offset calculation	None	uint32
224	E0	QEP_Alignment_Offset	MPOS-QEP	R	QEP offset angle	Degree	q16
225	E1	QEP_Linerization_Enable	QEP Calibration	W	Command to enable QEP calibration	None	uint32
226	E2	QEP_Main_Align_Offset	QEP Calibration	R/W	User input on QEP offset angle	Degree	q16
227	E3	fw_pid_Kp	Field Weakening	R/W	Kp of the PI controller	None	q16
228	E4	fw_pid_Ki	Field Weakening	R/W	Ki of the PI controller	None	q16
229	E5	LBG1_fc_bemf_mag	MPOS-Estimator	R/W	Cutoff frequency for back EMF low pass filter	Hz	q16
230	E6	Power_Measure_Iphase_Offset	Power Measurement	R/W	Current offset for power measurement	None	q14
231	E7	Power_Measure_Iphase_Factor	Power Measurement	R/W	Current gain for power measurement	None	q14
232	E8	id_pi_bandwidth	Obsolete	R			
233	E9	iq_pi_bandwidth	Obsolete	R			
234	EA	fw_vmax_scl_factor	Field Weakening	R/W	Voltage magnitude reference	None	q14
235	EB	hall_mech_angle_offset	MPOS-Hall	R/W	HALL signal offset	Degree	q16
236	EC	halledge_pol_skew_time	MPOS-Hall	R/W	HALL signal edge rate	msec	q16

237	ED	sys_fault	Status	R	Motor terminal faults	None	uint32
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